Indian Crime Analytics & Crime Prediction System

Project Type

Data Science + Machine Learning + Business Intelligence

Duration

1 Month / 4 Weeks

Tools & Technologies

Tool	Purpose
Python	Data cleaning, EDA, visualization, ML
Pandas & NumPy	Data processing
Matplotlib / Seaborn	Data visualization
Scikit-learn	Machine Learning
SQL / MySQL	Database & analytical queries
Excel	Initial data cleaning &

	reporting
Power BI	Interactive Crime Dashboard
Jupyter Notebook / VS Code	Development

For the project, students can use **NCRB/Open Government Data** as the real-world data source. The Government's data portal contains NCRB crime datasets, including state/UT-wise, city-wise and crime-category data.

🎯 Project Objective

The main objective is to build a system that can:

1. Analyze crime patterns across Indian states and cities.
2. Identify high-crime states/districts.
3. Analyze crime trends year-wise.
4. Find the most common crime categories.
5. Analyze crime against women, children and senior citizens.
6. Identify seasonal/monthly crime patterns where suitable data exists.
7. Use Machine Learning to **predict crime counts / crime-risk category**.
8. Store processed data in SQL.
9. Create an interactive **Power BI Crime Intelligence Dashboard**.
10. Generate useful insights for police administration and decision-making.

> **Important:** ML predictions should be presented as analytical estimates, **not as predictions of who will commit a crime or accusations against individuals.**

📊 Dataset Structure

A good internship version should contain multiple tables rather than one simple CSV.

1. `crime_cases`

Column	Example
Case_ID	C00001
Year	2023
State	Maharashtra
District	Pune
City	Pune
Crime_Type	Theft
Cases_Reported	1250
Cases_Solved	980
Cases_Pending	270

2. `crime_category`

Crime_ID	Crime_Type	Category
CR001	Murder	Violent
CR002	Theft	Property
CR003	Cyber Crime	Cyber

CR004	Kidnapping	Violent
CR005	Fraud	Economic

3. `state_population`

State	Year	Population
Maharashtra	2021	...
Karnataka	2021	...
Gujarat	2021	...

This allows students to calculate:

Crime Rate = Crime Cases / Population × 100,000

4. `police_resources`

State	Year	Police_Stations	Police_Strength
Maharashtra	2023	...	...

5. `crime_against_women`

Year	State	Crime_Type	Cases

2023	Maharashtra	Rape	...
2023	Maharashtra	Kidnapping	...

6. `cyber_crime`

Year	State	Cyber_Crime_Type	Cases
2023	Maharashtra	Online Fraud	...
2023	Karnataka	Identity Theft	...

🤖 Machine Learning Module

Make the ML component slightly advanced so that it looks like an actual Data Science internship project.

ML Problem 1 — Crime Count Prediction

Input Features

```
Year
State
District
Crime_Type
Population
Previous_Year_Cases
Police_Strength
Urbanization_Rate
```

Target

```
Crime_Cases
```

Possible algorithms:

- Linear Regression
- Random Forest Regressor
- Decision Tree Regressor
- Gradient Boosting Regressor

Compare models using:

```
MAE
RMSE
R² Score
```

🧠 ML Problem 2 — Crime Risk Classification

Create categories:

```
Low Risk
Medium Risk
High Risk
```

For example:

```
Crime Rate < threshold → Low
Crime Rate between thresholds → Medium
Crime Rate > threshold → High
```

Then train:

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost (optional)

Output

```
State: Maharashtra
District: Pune

Predicted Risk: HIGH
```

Again, this should be interpreted as **area-level analytical risk**, not individual criminal prediction.

🔍 Python EDA

Students should perform:

Univariate Analysis

```
Which crime occurs most frequently?
Which state has the highest number of reported cases?
```

Bivariate Analysis

```
Year vs Crime Cases
State vs Crime Cases
Crime Type vs Cases
Population vs Crime Rate
```

Trend Analysis

```
2019 → 2020 → 2021 → 2022 → 2023
```

Correlation Analysis

Study relationships between:

```
Population
Police Strength
Crime Rate
Crime Cases
```

🗄️ SQL Module

Import cleaned data into MySQL.

Students should write at least **25–30 SQL queries**.

Basic

```
SELECT *
FROM crime_cases;
```

Total Crimes

```
SELECT SUM(Cases_Reported) AS Total_Crimes
FROM crime_cases;
```

State-wise Crimes

```
SELECT State,
       SUM(Cases_Reported) AS Total_Crimes
FROM crime_cases
GROUP BY State
ORDER BY Total_Crimes DESC;
```

Top 10 Crime-Prone Districts

```
SELECT District,
       SUM(Cases_Reported) AS Total_Crimes
FROM crime_cases
GROUP BY District
ORDER BY Total_Crimes DESC
LIMIT 10;
```

Year-wise Trend

```
SELECT Year,
       SUM(Cases_Reported) AS Total_Crimes
FROM crime_cases
GROUP BY Year
ORDER BY Year;
```

Crime Category Analysis

```
SELECT c.Category,
       SUM(cr.Cases_Reported) AS Total_Cases
FROM crime_cases cr
JOIN crime_category c
ON cr.Crime_Type = c.Crime_Type
GROUP BY c.Category
ORDER BY Total_Cases DESC;
```

📈 Power BI Dashboard

Create **4 dashboards/pages**.

Dashboard 1 — India Crime Overview

KPIs:

```
Total Crimes
Total States
Total Districts
Most Reported Crime
Highest Crime State
Crime Growth %
```

Visuals:

- India Map
- Year-wise Line Chart
- Crime Category Donut Chart
- State-wise Bar Chart
- KPI Cards

👩 Crime Against Women Dashboard

KPIs:

```
Total Cases
Cases Solved
Cases Pending
Highest Crime State
Most Common Crime
```

Charts:

- State-wise crime
- Year-wise trend
- Crime type analysis
- State ranking
- Category distribution

💻 Cyber Crime Dashboard

Analyze:

```
Cyber Crime Cases
Online Fraud
Identity Theft
Financial Fraud
State-wise Cyber Crime
Year-wise Cyber Crime Growth
```

This gives the project a **modern Data Science + Cyber Analytics** component.

🗺️ Crime Risk Dashboard

Power BI should display:

```
State
District
Crime Rate
Previous Crime Rate
Predicted Crime
Risk Level
```

Example:

State	District	Crime Rate	Predicted Cases	Risk
Maharashtra	Pune	High	1450	High
Karnataka	Mysuru	Medium	720	Medium
Gujarat	Surat	Low	410	Low

Excel Module

Excel should not just be used for storing the CSV.

Students should perform:

Data Cleaning

- Remove duplicates
- Handle missing values
- Standardize state names
- Standardize crime names
- Change data types

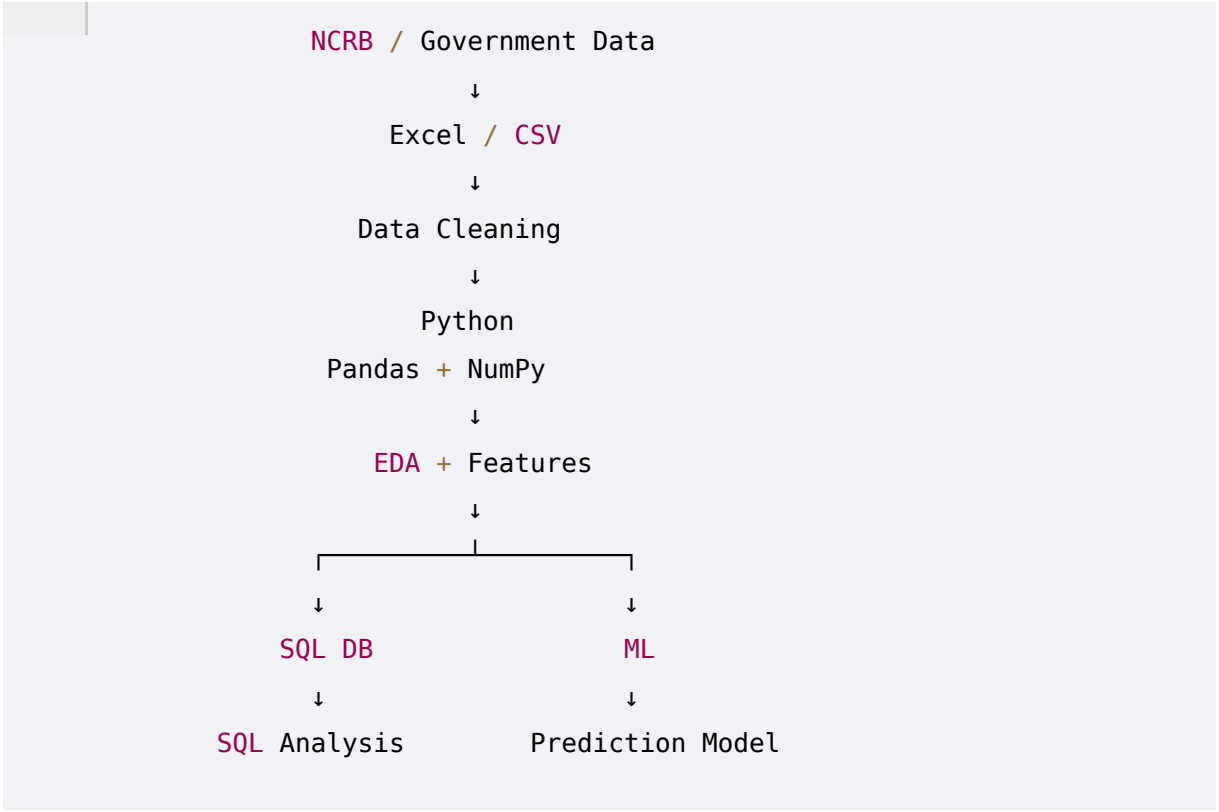
Excel Analysis

Use:

- Pivot Tables
- Pivot Charts
- XLOOKUP
- IF
- SUMIFS
- COUNTIFS
- Percentage calculations
- Conditional Formatting

Create an **Excel Crime Summary Report**.

Final Project Architecture



Recommended Project Folder

```
Indian_Crime_Analytics/
│
├── data/
│   ├── raw/
│   ├── cleaned/
│   └── processed/
│
├── excel/
│   └── crime_analysis.xlsx
│
├── sql/
│   ├── database.sql
│   └── queries.sql
│
├── python/
│   ├── data_cleaning.py
│   ├── eda.py
│   └── feature_engineering.py
```

Final Internship Deliverables

At the end, student should submit:

1. **Raw NCRB/Open Government dataset**
2. **Cleaned Excel dataset**
3. **Python EDA notebook**
4. **Python ML notebook**
5. **Trained ML model**
6. **MySQL database**
7. **30+ SQL queries**
8. **Power BI `.pbix` dashboard**
9. **Excel analytical report**
10. **Project documentation**
11. **PPT presentation**
12. **GitHub repository**
13. **Final viva/demo**

Strong Project Name for Resume

Indian Crime Analytics & Prediction System — Python, Machine Learning, SQL, Excel & Power BI

Resume description:

> Developed an end-to-end Indian Crime Analytics system using Python, SQL, Machine Learning, Excel and Power BI to analyze crime trends, state/district-level patterns and crime-risk categories. Built predictive models for crime-count estimation, performed data preprocessing and EDA, designed SQL analytical queries, and developed interactive Power BI dashboards for data-driven insights.

For legal terminology, the project can also acknowledge India's newer criminal-law framework: NCRB's official materials provide the **Bharatiya Nyaya Sanhita (BNS), 2023**, along with BNSS and BSA resources.